

# KUSHAGRA SINHA

SOFTWARE DEVELOPMENT ENGINEER | BACKEND & DISTRIBUTED SYSTEMS | PYTHON | AI / LLM SYSTEMS

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## EXPERIENCE

### Software Engineer Intern

Aug 2025 - Dec 2025

#### Tesla

Austin, TX

- Built a **natural language dashboard configuration platform** using Python, FastAPI, Elasticsearch, and LangGraph that translated natural language queries into validated Elasticsearch DSL through deterministic templates and a human-in-the-loop confirmation step, reducing dashboard widget creation time by **90%**.
- Developed a **dataset onboarding service** that transformed Excel based specifications into validated JSON schemas using FastAPI, Pydantic, LLM and concurrent processing, reducing onboarding time from **several days to under an hour**.
- Engineered a **conversational search system** using Python, Elasticsearch, Redis, and LangGraph that performed retrieval across assets, cases, and ticket datasets while maintaining session context across interactions, improving retrieval accuracy by **80%**.
- Built a **retrieval augmented email generation API** using Python, ClickHouse vector search, and embeddings to automatically draft context aware emails from historical cases, user data, and internal knowledge sources, **reducing draft creation time** and ensuring consistent, relevant content.
- Designed an **idempotent ticket creation agent** integrated with Microsoft Teams that verified ticket creation through backend API confirmation before responding to users, preventing duplicate ticket creation and **eliminating hallucinated confirmations**.
- Tech Stack:** Python, FastAPI, LangGraph, Elasticsearch, Redis, ClickHouse, Pydantic, RAG, Vector Search

### Software Engineer

Feb 2023 - Aug 2024

#### Qualcomm

Bangalore

- Built a **multithreaded XML ETL pipeline** using Python, NumPy, MongoDB, and Elasticsearch to process hundreds of 8–9 GB GPU capture files, reducing processing time from **weeks to four days** and enabling the graphics driver team to query capture data.
- Replaced a **FIFO CI/CD workflow with a RabbitMQ** based priority scheduler integrated with Jenkins that automatically promoted urgent release builds, eliminating manual queue intervention by engineering teams.
- Tech Stack:** Python, NumPy, MongoDB, Elasticsearch, RabbitMQ, Jenkins, Multithreading

### Software Engineer

Feb 2022 - Dec 2022

#### Zomato

Gurugram

- Built an **event driven inventory alerting service** using Go, Kafka, Redis, and MySQL that consumed demand forecasts and generated automated restocking alerts for kitchen administrators, **reducing stock outs** across cloud kitchens.
- Owned **backend payment integrations for Zomato Gold** using Go, DynamoDB, gRPC, and Protobuf, implementing retries, circuit breakers, and idempotent writes to support more than 100,000 daily transactions under production load.
- Developed **backend APIs for the Zomato Instant Admin platform** secured with JWT authentication and role based access control, enabling sales teams to **onboard cloud kitchens, manage menus, and update pricing** without engineer operated SQL workflows.
- Tech Stack:** Golang, gRPC, Protobuf, Apache Kafka, Redis, DynamoDB, MySQL, JWT

### Software Engineer

Feb 2021 - Dec 2021

#### Bluestone

Bangalore

- Engineered **catalog and pricing pipelines** using Java and PostgreSQL with transactional rollback and idempotent batch processing across 10,000+ SKUs, preventing partial catalog updates and ensuring data consistency.
- Tech Stack:** Java, PostgreSQL, SQL

## PROJECTS

### RolloutX - Distributed Feature Flag Platform

[github.com/sinhakushagra21/RolloutX](https://github.com/sinhakushagra21/RolloutX)

May 2026 - Jun 2026

- Built a **distributed feature flag platform** with separate control and data planes, where a transactional outbox publishes immutable versioned snapshots to Kafka and stateless data-plane replicas evaluate flags in-process over lock-free snapshots at **211 ns** per evaluation (~4.7M evaluations/sec per core).
- Implemented deterministic, sticky percentage rollouts with MurmurHash3 bucketing, property-tested to hold the distribution within **±1% across 1,000,000 keys**, with sub-second kill switches and a last-known-good SDK fallback so flags keep evaluating during outages.
- Tech Stack:** Go, gRPC, PostgreSQL, Kafka, Redis, Prometheus, OpenTelemetry, Kubernetes, Terraform

### Stateful Multi-Agent Fitness Planner

[github.com/sinhakushagra21/fitgenai](https://github.com/sinhakushagra21/fitgenai)

Jan 2026 - Apr 2026

- Built a **stateful multi-agent fitness AI coach** with an LLM router that directs each turn to specialist diet and workout tools, vector search RAG over saved plans, and a plan evaluator with a refinement loop plus scope guardrails, backed by 220+ tests.
- Tech Stack :** Python, LangGraph, OpenAI API, MongoDB Atlas, Redis, RAG, LangSmith

### Fast Commerce Database Platform

[github.com/sinhakushagra21/fast-commerce-sql-dmdd](https://github.com/sinhakushagra21/fast-commerce-sql-dmdd)

Sep 2024 - Dec 2024

- Designed a **normalized relational database schema** supporting customers, orders, inventory, delivery operations, and payment workflows for a fast grocery delivery platform.
- Implemented stored procedures, reporting views, delivery-slot validation logic, and role-based access controls using PL/SQL.
- Tech Stack:** Oracle, PL/SQL, SQL

## TECHNICAL SKILLS

|                          |                                                                                                                                       |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Languages:               | Python, Go, Java, SQL, JavaScript, C++, PL/SQL                                                                                        |
| Frameworks:              | FastAPI, Flask, Django, Spring Boot, Gin                                                                                              |
| APIs & Protocols:        | REST APIs, gRPC, Protocol Buffers (Protobuf), OAuth, JWT                                                                              |
| Distributed Systems:     | Apache Kafka, RabbitMQ, Microservices, Event-Driven Systems, Concurrency, Multithreading                                              |
| Databases:               | Redis, Elasticsearch, PostgreSQL, MySQL, MongoDB, DynamoDB, Cassandra, Oracle Database                                                |
| Cloud & DevOps:          | AWS, Docker, Kubernetes, Terraform, Jenkins, GitHub Actions, Linux                                                                    |
| AI Assisted Development: | LangGraph, LangChain, Retrieval-Augmented Generation (RAG), Vector Search, Pinecone, FAISS, LangSmith, GitHub Copilot, Cursor, Claude |

## EDUCATION

### Northeastern University — M.S., Software Engineering Systems

Aug 2024 - Jul 2026

### PES University — B.Tech, Electronics & Communication Engineering

Aug 2016 - Aug 2020